

CSE 219 Practice Questions

FFT, Sampling, Laplace Transform, and z-Transform

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1. Radix-2 DIT Cooley-Tukey FFT

Problem 1.1. Starting from $X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn}$, derive the radix-2 DIT split into even and odd subsequences and the butterfly relations for $0 \leq k < N/2$.

Problem 1.2. Compute the 4-point DIT FFT of $x[n] = [2, 1, -3, 4]$ using radix-2 butterflies.

Problem 1.3. Compute the 4-point DIT FFT of $x[n] = [1, 2, 3, 4]$ using radix-2 butterflies.

Problem 1.4. Compute the 8-point DIT FFT of $x[n] = [1, 2, 0, -1, 3, 0, 2, 1]$ using the full radix-2 butterfly.

Problem 1.5. Recover $x[n]$ from

$$X[k] = [6, \sqrt{2} - j\sqrt{2}, 2 + 4j, -\sqrt{2} - j\sqrt{2}, -2, -\sqrt{2} + j\sqrt{2}, 2 - 4j, \sqrt{2} + j\sqrt{2}]$$

using an 8-point inverse FFT.

2. 2D Matrix Formulation of Cooley-Tukey (Bailey's FFT)

Problem 2.1. Show how a 4-point DFT can be written as a 2×2 matrix factorization using column-major input, row DFTs, twiddles, and column DFTs.

Problem 2.2. Use a 2×4 Bailey factorization to compute the 8-point DFT of $x[n] = [1, 3, -1, 2, 0, 4, 2, 1]$.

Problem 2.3. A 16-point sequence is $x[n] = [1, 0, 0, 0, 2, 0, 0, 0, -1, 0, 0, 0, 3, 0, 0, 0]$. Use the 4×4 formulation to reduce the work.

3. Bluestein's FFT

Problem 3.1. Derive Bluestein's chirp identity

$$W_N^{nk} = W_{2N}^{-(k-n)^2} W_{2N}^{k^2} W_{2N}^{n^2}.$$

Problem 3.2. Use Bluestein's construction to compute the 3-point DFT of $x[n] = [2, -1, 3]$.

Problem 3.3. Run the same method on the 4-point sequence $x[n] = [1, 2, 0, 0]$ and compare with the ordinary 4-point DFT.

4. Application of FFT: Fast Convolution

Problem 4.1. Compute the linear convolution of $x[n] = [1, 2]$ and $h[n] = [1, -1]$ using a 4-point FFT.

Problem 4.2. Compute the linear convolution of $x[n] = [1, -1, 2]$ and $h[n] = [2, 1]$ using a 4-point radix-2 FFT.

Problem 4.3. Compute the linear convolution of $x[n] = [1, 0, 1]$ and $h[n] = [1, -1, 2]$. Use the smallest radix-2 FFT size.

Problem 4.4. Repeat Problem 4.3 incorrectly with a 4-point FFT and show the circular-convolution wrap-around.

5. Nyquist-Shannon Sampling Theorem

Problem 5.1. State the Nyquist-Shannon sampling theorem in both angular-frequency and ordinary-frequency form.

Problem 5.2. Derive the condition $\omega_s > 2\omega_M$ by comparing adjacent spectral copies of a bandlimited spectrum.

Problem 5.3. A lowpass signal is bandlimited to $|f| \leq 1.6$ kHz. Sketch the sampled spectrum for $f_s = 4.5$ kHz and for $f_s = 2.8$ kHz, and identify which case aliases.

Problem 5.4. A signal contains tones at 1.2 kHz and 1.8 kHz. Give one safe sampling frequency and one unsafe sampling frequency.

6. Ideal Reconstruction, Zero-Order Hold, Linear Interpolation

Problem 6.1. Write the impulse response and frequency response of the ideal reconstruction filter, and decide whether it is causal.

Problem 6.2. Derive the impulse response and frequency response of the zero-order hold (ZOH).

Problem 6.3. Derive the impulse response and frequency response of linear interpolation.

Problem 6.4. Compare the magnitude responses of ideal, ZOH, and linear interpolation at $\omega = \omega_s/4$.

Problem 6.5. State one main advantage and one main drawback of each of the three reconstruction methods.

7. Laplace Transform: Definition, Properties, ROC, Poles and Zeros

Problem 7.1. Find the Laplace transform and ROC of $x(t) = (t - 2)e^{-3(t-2)}u(t - 2)$.

Problem 7.2. Find the Laplace transform and ROC of $x(t) = t^2 e^{-3t}u(t)$.

Problem 7.3. Find the Laplace transform and ROC of $x(t) = te^{-t}u(t) + e^{2t}u(-t)$.

Problem 7.4. Find the Laplace transform and ROC of $x(t) = e^{-4(t-1)}\cos(3(t-1))u(t-1)$.

Problem 7.5. Find the Laplace transform, ROC, poles, and zeros of $x(t) = te^{-t}\sin(2t)u(t)$.

Problem 7.6. Find the Laplace transform and ROC of $y(t) = (e^{-t}\sin(3t)u(t)) * (e^{-2t}u(t))$.

8. Inverse Laplace Transform

Problem 8.1. Invert $X(s) = \frac{3s+7}{(s+1)(s+2)}$ with ROC $\text{Re}\{s\} > -1$.

Problem 8.2. Invert $X(s) = \frac{2s+3}{s^2+2s+5}$ with ROC $\text{Re}\{s\} > -1$.

Problem 8.3. Invert $X(s) = \frac{-5}{(s+1)(s-4)}$ with ROC $-1 < \text{Re}\{s\} < 4$.

Problem 8.4. Invert $X(s) = \frac{s}{(s^2+4)^2}$ with ROC $\text{Re}\{s\} > 0$.

Problem 8.5. Invert $X(s) = \frac{e^{-s}}{s(s+2)}$ with ROC $\text{Re}\{s\} > 0$.

Problem 8.6. Invert $X(s) = \frac{2}{s(s^2+4)}$ with ROC $\text{Re}\{s\} > 0$.

Problem 8.7. Invert $X(s) = \frac{s}{s+2}$ with ROC $\text{Re}\{s\} > -2$.

9. Applications of Laplace Transform

Problem 9.1. For $H(s) = \frac{s+2}{(s+1)(s+2)}$, determine the admissible ROC choices and the corresponding causality/stability properties.

Problem 9.2. For $H(s) = \frac{1}{(s+1)^2(s+3)}$, determine the admissible ROC choices and the corresponding causality/stability properties.

Problem 9.3. Decide whether the Fourier transform exists for $X(s) = \frac{1 - e^{-3(s+2)}}{s+2}$. Interpret the result in time domain.

Problem 9.4. Solve $y'(t) + y(t) = e^{-2(t-1)}u(t-1)$ with $y(0^-) = 0$.

Problem 9.5. Solve $y''(t) + 3y'(t) + 2y(t) = e^{-(t-1)}u(t-1)$ with zero initial conditions.

Problem 9.6. A causal LTI system is described by the LCCDE $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y(t) = \frac{dx}{dt} + 3x(t)$. Find $H(s)$, sketch the pole-zero plot, determine whether the system is stable, and find the impulse response $h(t)$.

10. z-Transform: Definition, Properties, ROC, Poles and Zeros

Problem 10.1. Find the z-transform, poles, zeros, and ROC of $x[n] = \left(\frac{3}{4}\right)^n \cos\left(\frac{\pi n}{3}\right)u[n]$.

Problem 10.2. Find the z-transform and ROC of $x[n] = (n-2)\left(\frac{1}{3}\right)^{n-2}u[n-2]$.

Problem 10.3. Find the z-transform, poles, zeros, and ROC of $x[n] = \left(\frac{1}{4}\right)^n u[n] - 2^n u[-n-1]$.

Problem 10.4. Find the z-transform and zero pattern of $x[n] = \delta[n] - 2\delta[n-1] + 2\delta[n-2] - \delta[n-3]$.

Problem 10.5. Find the z-transform, poles, zeros, and ROC of $x[n] = \left(\frac{1}{2}\right)^{n-1} \cos\left(\frac{\pi(n-1)}{3}\right)u[n-1]$.

Problem 10.6. Find the z-transform, poles, zeros, and ROC of $y[n] = \left(\left(\frac{1}{2}\right)^n u[n]\right) * \left(\left(\frac{1}{3}\right)^n u[n]\right)$.

Problem 10.7. Let $x[n] = \left(\frac{1}{2}\right)^n u[n]$. Find the z-transform and ROC of the running sum $y[n] = \sum_{k=-\infty}^n x[k]$.

11. Inverse z-Transform

Problem 11.1. Invert $X(z) = \frac{2 - z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{3}{4}z^{-1})}$ for $|z| > \frac{3}{4}$.

Problem 11.2. Invert $X(z) = \frac{1}{(1 - 2z^{-1})(1 - \frac{1}{3}z^{-1})}$ for $\frac{1}{3} < |z| < 2$.

Problem 11.3. Invert $X(z) = \frac{1 - \frac{5}{6}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}$ for $|z| < \frac{1}{3}$.

Problem 11.4. Invert $X(z) = \frac{1 + \frac{1}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})^2}$ for $|z| > \frac{1}{2}$.

Problem 11.5. Invert $X(z) = z^{-1} \frac{1 - \frac{1}{4}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$ for $|z| > \frac{1}{2}$.

12. Applications of z-Transform: Causality, Stability from ROC

Problem 12.1. For $H(z) = \frac{1 - 2z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}$, determine the valid ROC choices and the corresponding causality/stability/DTFT conclusions.

Problem 12.2. For $H(z) = \frac{1}{(1 - 0.8z^{-1})(1 - 1.25z^{-1})}$, determine whether any realization can be both stable and causal.

Problem 12.3. Determine whether the DTFT exists for $x[n] = (\frac{1}{2})^n u[n] - 1.2^n u[-n - 1]$.

Problem 12.4. Given $h[n] = (\frac{2}{3})^n u[n] - (\frac{4}{3})^n u[-n - 1]$, find $H(z)$ and its ROC, and determine whether the system is causal, stable, and whether the DTFT of $h[n]$ exists.

Problem 12.5. A system has poles at $0.95e^{\pm j\pi/4}$ and 1.1. Determine whether there exists a stable realization, a causal realization, and a realization that is both.